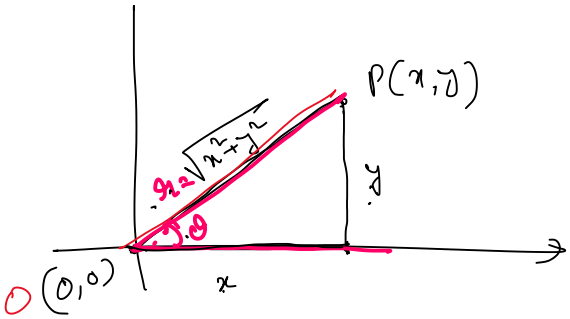
Lecture 17

Wednesday, 16 September 2026 8:26 AM

Recap: $\int_a^b f(x)\,dx = \sup(\text{lower sum}) = \inf(\text{upper sum})$

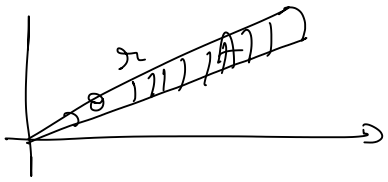
Today: Sketching graph of eqn like $(x^2+y^2)^3 = y^2$.



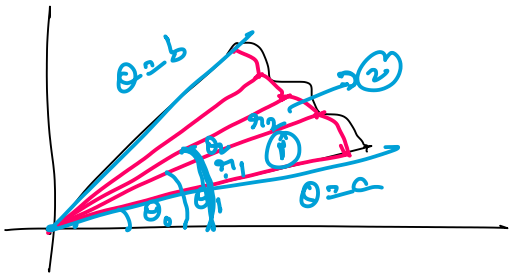
Rectangular Coord system	Polar Coordinate system
$P(x,y)$	$P(r,\theta)$

$x = r\cos\theta$

$y = r\sin\theta$



Area of sector $= \frac{1}{2} r^2\theta$



Area of sector 1 $= \frac{1}{2} r_1^2(\theta_1-\theta_0)$

Area of sector 2 $= \frac{1}{2} r_2^2(\theta_2-\theta_1)$

⋮

Area of the region $= \frac{1}{2} r_1^2(\theta_p-\theta_0)$

$+\frac{1}{2} r_2(\theta_2 - \theta_1)$

$+ \cdots$

$r = f(\theta)$

Area of the sectorial type region

$$= \frac{1}{2}\int_{\theta=a}^{b} (f(\theta))^2 \, d\theta$$

$x = r\cos\theta$,

$y = r\sin\theta$

θ = constants $\longrightarrow$ $x = y$ or line eqn

r = constant $\longrightarrow$ circle

$x = 2\cos\theta$

$y = 2\sin\theta$

$x^2 + y^2 = 4$

Eg 1: Sketch the graph of $(x^2+y^2)^3 = y^2$ and find the area enclosed by it.

$x = r\cos\theta$, $y = r\sin\theta$

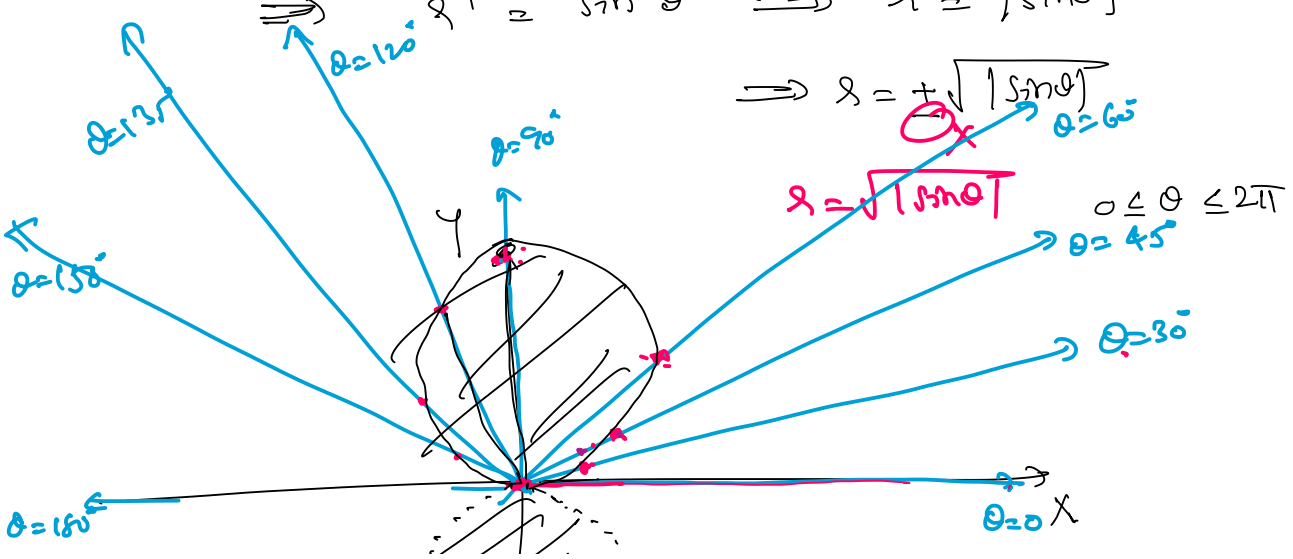
Eqn $\Rightarrow$ $r^6 = r^2\sin^2\theta$

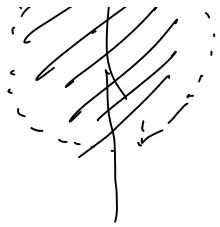
$\Rightarrow$ $r^4 = \sin^2\theta \Rightarrow r^2 = |\sin\theta|$

$\Rightarrow r = \pm\sqrt{|\sin\theta|}$

$r = \sqrt{|\sin\theta|}$

$0 \le \theta \le 2\pi$





$r = f(\theta) = \sqrt{|\sin\theta|}$

$0 \le \theta \le 2\pi$

$r(\theta) = r(-\theta)$

$\theta = 0$

$x = r\cos\theta$

$y = r\sin\theta$

$x = r, r > 0$

$y = 0$

$$\text{Area} = \frac{1}{2}\int_{\theta=0}^{2\pi} f^2(\theta)\, d\theta$$

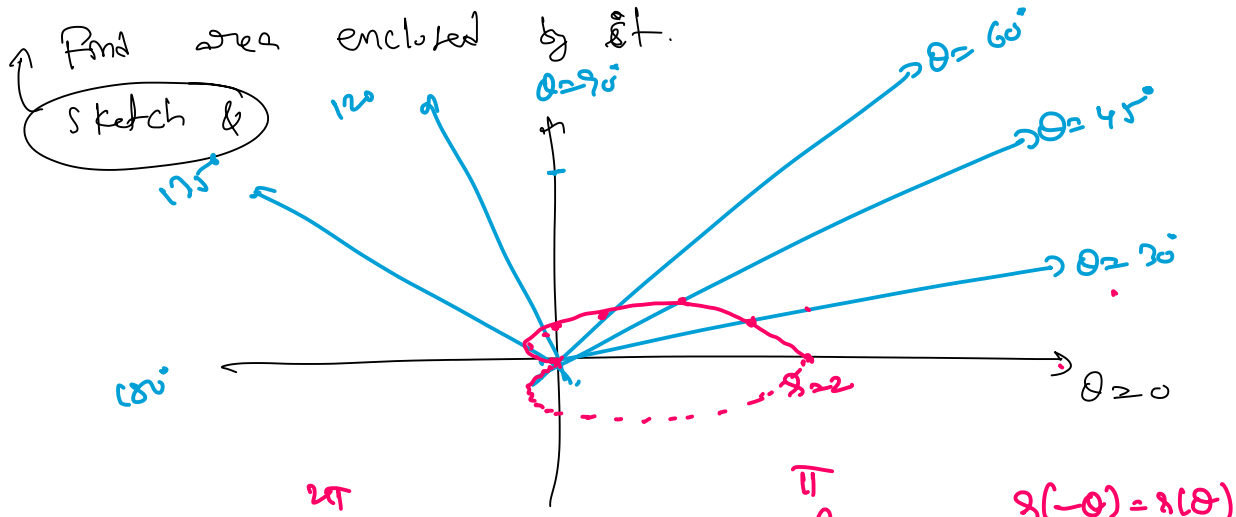
$$= \frac{1}{2}\int_{\theta=0}^{2\pi} \left(\sqrt{|\sin\theta|}\right)^2 d\theta$$

$$= \frac{1}{2} \times 4 \int_{\theta=0}^{\pi/2} \sin\theta\, d\theta = 2.\ (-\cos\theta)\Big|_{\theta=0}^{\pi/2} = 2$$

Eg): $r = 1 + \cos\theta, \quad 0 \le \theta \le 2\pi.$

Find area enclosed by it.

Sketch &



$$\text{Area} = \frac{1}{2}\int_{\theta=0}^{2\pi} (1+\cos\theta)^2\, d\theta = \frac{1}{2}\times 2\int_{\theta=0}^{\pi} (1+\cos\theta)^2\, d\theta \qquad r(-\theta) = r(\theta)$$

$$= 5\pi/2$$

Ex!: $r = f(\theta) = \sqrt{|\cos 2\theta|} \qquad 0 \le \theta \le 2\pi$